POWER & CHARGING

Keeping phones, lights, radios and medical devices running in an outage, with a worksheet to size your backup power.

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GUIDE / VERIFY: General preparedness information, not a substitute for professional training, local rules, manufacturer instructions or medical care. **Practice skills in safe conditions before you need them, and verify important information with current official sources.**

1. FIRST, NEED LESS POWER

- Turn on your phone's **low-power / battery-saver mode** the moment the power goes out.
- Lower screen brightness; close apps; turn off Bluetooth and Wi-Fi when you're not using them.
- Text instead of calling. Texts use less battery and get through busy networks more easily.
- Use one phone at a time for news and keep the others off or in airplane mode as backups.
- Use LED lights; they run many times longer on the same batteries.
- A battery or crank radio gets news without using your phone. *(TR-08)*

2. BACKUP POWER OPTIONS

OPTION	GOOD FOR	KNOW THIS
Spare batteries	Flashlights, radios, lanterns.	Standardize on one or two sizes (AA, AAA). Lithium AA/AAA last longest in storage and in cold. Check package dates.
Power bank	Phones, small lights, radios with USB charging.	Keep charged; top up every 3 months. See the worksheet for how many charges you'll get.
Car	Charging phones and power banks with a car charger.	**Only with the car outdoors, never in a garage**, even with the door open. Run the engine so you don't drain the battery.
Portable power station	Phones, laptops, lights, CPAP machines, a small fridge for a while. Silent; safe indoors.	Rated in watt-hours (Wh). Recharge from the wall, a car or solar panels.
Solar panel	Recharging power banks and power stations over several days.	Output depends on sun and angle. Expect a panel's watts x about 3 to 5 hours of good sun per day, minus losses.
Generator	Fridge, freezer, sump pump, well pump, furnace fan.	Outdoors only, **at least 20 feet** from windows, doors and vents. See the warning below.

(!) **GENERATORS KILL PEOPLE EVERY YEAR.** Carbon monoxide: never run one in a home, garage, basement, crawlspace or shed, or near an open window. Have battery CO alarms inside. **Never plug a generator into a wall outlet** ("backfeeding"): it can electrocute utility workers and neighbors. Use heavy-duty outdoor extension cords, or a transfer switch installed by a licensed electrician. Turn the generator off and let it cool before refueling.

3. THE MATH, IN PLAIN TERMS

- **Watts (W)** = how fast a device uses power. Usually printed on its label or power adapter.
- **Watt-hours (Wh)** = how much energy a battery holds. 100 Wh can run a 10 W device for about 10 hours.
- **Power banks** are often labeled in milliamp-hours (mAh). To convert: **mAh x 3.7 ÷ 1,000 = Wh**.
- You never get the full rating out. Plan on about **two-thirds** of a power bank's rating reaching your phone, and about 85% of a power station's.
- A typical phone battery holds about 12 to 20 Wh. A 20,000 mAh power bank (74 Wh) charges most phones about 3 times.
- Motors (fridges, pumps) take several times their running watts for a moment when they start. Size generators and inverters for that.

4. BACKUP POWER WORKSHEET

For each device: watts x hours per day = watt-hours per day. Add them up, multiply by days, then add about 20% for losses.

DEVICE	WATTS	HOURS PER DAY	WH PER DAY	EXAMPLE
				Phone (2 charges/day): 30 Wh
				LED lantern: 5 W x 5 h = 25 Wh
				Radio: 2 W x 6 h = 12 Wh
				CPAP: 40 W x 8 h = 320 Wh*
				Laptop: 50 W x 2 h = 100 Wh
TOTAL WH PER DAY				487 Wh
x DAYS x 1.2 = WH NEEDED				3 days: 1,753 Wh

** CPAP use varies widely (heated humidifier uses much more). Check your machine's manual or ask the supplier for its battery runtime.*

5. MEDICAL DEVICES THAT NEED POWER

- Ask your doctor or equipment supplier how long the device runs on battery, and what backup they recommend.
- Keep spare batteries charged, and the charger with the device.
- Tell your power company: many have a medical-needs list for priority information and help.
- Medicines that need refrigeration: ask your pharmacist how long they last unrefrigerated, and keep a cooler and ice packs.
- Know where to go (a hospital, a shelter with power) if the outage will outlast your battery.

SOURCES / FURTHER REFERENCE

1. FEMA: Ready.gov: Power Outages; Individuals with Disabilities (power-dependent equipment). ready.gov/power-outages
2. Centers for Disease Control and Prevention: Avoiding Carbon Monoxide Poisoning (generators). cdc.gov/carbon-monoxide
3. U.S. Consumer Product Safety Commission: Portable Generator Safety. cpsc.gov
4. U.S. Department of Energy: Energy Saver: appliance energy use. energy.gov/energysaver

IMPORTANT: Device wattages vary; use the numbers on your own equipment. Follow the manufacturer's instructions for every generator, battery and charger.